Dairy Campus Research Data Platform

Data Descriptor, Standardised Metadata Dictionaries and Calculation Methodology

Bert Klandermans Fedde de Jong

Dairy Campus, Wageningen University & Research, Leeuwarden, The Netherlands

DOI: 10.5281/zenodo.24092174 | Licence: CC BY 4.0

Abstract

Dairy Campus (Wageningen University & Research, Leeuwarden) maintains an experimental dairy herd of approximately 500 Holstein Friesian dairy cows tracked continuously since 2016. Spanning more than a decade of high-throughput precision livestock farming observations, the platform encompasses millions of individual milking records, load-cell roughage intake visits, enteric greenhouse gas fluxes (CH_4, CO_2), and laboratory milk analyses. This data descriptor formalizes the complete research data platform: 19 distinct datasets encompassing 507 scientific variables, structured as standardized distributions (`dct:distribution`) of a single central catalog under W3C DCAT and DataCite standards. Derived quantities computed in the SQL layer (roughage intake, water exclusion deduction, laboratory dry matter intake, yield-weighted milk composition, and lactation demographics) are specified in closed form.

Contents

1 Longitudinal experimental infrastructure (500 dairy cows, 2016–present) 3

2 Calculation methodology 3

2.1 Roughage intake per visit and per day . . . 3
2.2 Correction for water added in the feed kitchen . . . 3
2.3 Dry matter intake . . . 3
2.4 Yield-weighted milk composition . . . 4
2.5 Lactation and reproduction timing . . . 4

3 Dataset overview 4

4 Data dictionaries 5

4.1 Master daily integrated cow summary . . . 5
4.2 Automatic concentrate feeder transactions . . . 11
4.3 Roughage intake control (RIC) visit transactions . . . 12
4.4 Daily roughage intake and dry matter integration . . . 13
4.5 Trioliet Triomatic feeding robot loading records . . . 13
4.6 Trioliet Triomatic feeding robot trough deliveries . . . 14
4.7 GEA separation and smart gate passages . . . 15
4.8 GEA milking transactions . . . 15
4.9 CowConnect mixer wagon rations . . . 16
4.10 Nedap Velos behaviour intervals . . . 16
4.11 Nedap Velos daily behaviour budgets . . . 17

4.12 CowConnect feed distribution drops . . . 17
4.13 Laboratory dry matter content (drying oven) . . . 17
4.14 GreenFeed enteric emission summaries . . . 18
4.15 GreenFeed bait feed drops . . . 18
4.16 Dairy Campus meteorological station . . . 19
4.17 Data completeness audit matrix . . . 20
4.18 Barn environment and animal daily profile . . . 20
4.19 Experimental cohort animal daily records . . . 23

5 Data availability and citation 25

1 Longitudinal experimental infrastructure (500 dairy cows, 2016–present)

Dairy Campus operates a high-throughput, multi-sensor experimental dairy infrastructure tracking an active experimental herd of approximately 500 Holstein Friesian dairy cows continuously since 2016. Longitudinal data streams encompass:

- **GEA automated milking systems and parlors** – millions of individual milking records including session yields, duration, quarter electrical conductivity, and walkthrough body weights.
- **Hokofarm RIC troughs** – millions of individual roughage intake visits on high-precision load cells with RFID cow recognition.
- **Trioliet Triomatic robot** – automated kitchen ingredient batch mixing, loading, added water mass, and trough delivery drops.
- **Hokofarm/SAC concentrate stations** – individual programmed concentrate allocations (A–D) versus consumed amounts.
- **C-Lock GreenFeed hoods** – voluntary voluntary bait-fed visits measuring individual enteric CH_4 and CO_2 flux rates.
- **Nedap Velos smarttags** – continuous behavioral accelerometry (eating, ruminating, lying, walking).
- **Meteorological weather station** – ambient temperature, relative humidity, wind speed, solar radiation, and rain.

Laboratory drying-oven dry matter analyses and official milk recording (CRV/MPR) complement the sensor streams.

2 Calculation methodology

This section formalises the derived variables that are computed inside the SQL tables and views of the platform. Only calculations documented in the data dictionary are included; the variable names given in parentheses refer to the columns in Section 4. Time stamps are denoted t, masses M (kg), and the index d refers to a calendar day.

2.1 Roughage intake per visit and per day

The Hokofarm RIC troughs are mounted on load cells. The intake and duration of visit v, and the daily fresh intake of roughage fraction $f \in \{1, 2, 3, 4\}$ (`rv1.Prod` – `rv4.Prod`), are

$$I_v = M_v^{\text{start}} - M_v^{\text{end}}, \qquad \Delta t_v = t_v^{\text{end}} - t_v^{\text{start}}, \qquad I_{f,d} = \sum_{v \in \mathcal{V}_{f,d}} I_v, \tag{1}$$

where $\mathcal{V}_{f,d}$ is the set of visits of a cow to troughs holding fraction f on day d.

2.2 Correction for water added in the feed kitchen

Water is added to the ration in the Trioliet Triomatic feed kitchen. Intake excluding added water (`rv`f`.ProdExclWater`, `opnMinWater`; view `VL_Totaal_sub_ric_opn_prod`) is

$$I_{f,d}^{\text{excl}} \;=\; \frac{M_{\text{loaded}} - M_{\text{water}}}{M_{\text{loaded}}}\; I_{f,d}, \tag{2}$$

with M_{loaded} the total mass loaded into the mixer and M_{water} the mass of water added to that batch.

2.3 Dry matter intake

Weekly dry matter contents DM_f (%) are determined gravimetrically in the drying oven (`VL_totaal_droogstoof`). Total daily dry matter intake (`tot.opnDs`, kg DM/d) is

$$\text{DMI}_d \;=\; \sum_{f=1}^{4} I_{f,d}^{\text{excl}}\, \frac{\text{DM}_f}{100}. \tag{3}$$

2.4 Yield-weighted milk composition

Milk recording (CRV/MPR) concentrations are aggregated per ISO calendar week w as yield-weighted means (`mpr_week.*`; view `VL_Dier_Datum_Crv`):

$$\bar{c}_w = \frac{\sum_{j \in \mathcal{J}_w} c_j \, Y_j}{\sum_{j \in \mathcal{J}_w} Y_j}, \qquad c \in \{\text{fat, protein, lactose, urea, SCC}\}, \tag{4}$$

where $\mathcal{J}_w$ is the set of test milkings in week w, c_j the concentration and Y_j (kg) the milk yield of milking j.

2.5 Lactation and reproduction timing

For a record on date t (all quantities in days):

$$\text{DIM} = t - t_{\text{calving}} \qquad (\texttt{LacDgn}, \texttt{IMDim}, \texttt{DaysInLactation}) \tag{5}$$

$$A = t - t_{\text{birth}} \qquad (\texttt{AgeDays}) \tag{6}$$

$$t_{\text{exp. calving}} = t_{\text{insemination}} + 280 \qquad (\texttt{VrwKlfDat}, \texttt{ExpecCalvingDate}) \tag{7}$$

$$D_{\text{to calving}} = t_{\text{exp. calving}} - t \qquad (\texttt{VrwDgn}) \tag{8}$$

3 Dataset overview

Table 1 lists all datasets; names link to the corresponding data dictionary.

Table 1: Datasets of the Dairy Campus research data platform.

#	Dataset	Content	Variables
1	`TVL_total`	Master daily integrated cow summary	97
2	`Hokofarm_SAC_concentrates`	Automatic concentrate feeder transactions	42
3	`Hokofarm_RIC`	Roughage intake control (RIC) visit transactions	18
4	`TVL_RIC_intake_dm`	Daily roughage intake and dry matter integration	12
5	`Triomatic_RIC_load`	Trioliet Triomatic feeding robot loading records	46
6	`Triomatic_RIC_drop`	Trioliet Triomatic feeding robot trough deliveries	46
7	`GEA_gates`	GEA separation and smart gate passages	5
8	`GEA_milkings`	GEA milking transactions	13
9	`TMR_Trioliet_cowconnect_mix`	CowConnect mixer wagon rations	12
10	`Velos_behaviour`	Nedap Velos behaviour intervals	4
11	`Velos_behaviour_day`	Nedap Velos daily behaviour budgets	7
12	`TMR_Trioliet_cowconnect_drop`	CowConnect feed distribution drops	11
13	`DC_DM_content`	Laboratory dry matter content (drying oven)	16
14	`Greenfeed_summarized_data`	GreenFeed enteric emission summaries	24
15	`Greenfeed_feed_aggregrated`	GreenFeed bait feed drops	13
16	`Weatherstation`	Dairy Campus meteorological station	18
17	`Missing_data`	Data completeness audit matrix	17
18	`Environment_barn_Animal_day`	Barn environment and animal daily profile	52
19	`Environment_barn_Animal_experim`	Experimental cohort animal daily records	54

4 Data dictionaries

All dictionaries share an identical layout. *Origin* indicates whether a variable is measured by a sensor or laboratory, or derived in the SQL layer; derivations refer to the equations in Section 2. A dash (—) denotes information that is not documented.

4.1 Master daily integrated cow summary

Dataset: `TVL_total` | **Source table:** `tvl_totaal` | **Variables:** 97

Table 2: Data dictionary of `TVL_total`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AnimalServerId`	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
`AntMlkA`	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`AntMlkEtm`	integer	–	Measured	Amount of milkings in the last 24 hours. 2 is the assumed value. 1 or 3+ means something is off during morning and/or afternoon milking.	—
`AntMlkO`	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
`bcs`	integer	–	Measured	Automatic body condition score camera	—
`BegDatPrf`	date	yy-mm-dd	Measured	Expected start date of the experiment	—
`BehCod`	string	text	—	Abbreviaton of the treatname name. Set by the researcher.	—
`BehNam`	string	text	—	Full name of the treatment. Set by the researcher	—
`Blk`	integer	–	—	Number of the assigned block. Set by the researcher.	—
`C21-10`	number	kg	Derived	Dag totaal krachtvoer caroussel	—
`C21-9`	number	kg	—	Kilogram of concentrate feed dispatched at the milking caroussel. NOT the uptake, that cannot be registered currently.	—
`CelStd`	string	x1000 cells/mL	Derived	Somatic cell count of the monthly milk test	Official somatic cell count (SCC) determined from routine 4-weekly milk recording samples (MPR), analyzed by laboratory. *Source:* `T_Animal_Milk Samples (MlkCon. SCCDay)`
`Dat`	date	yy-mm-dd	Measured	Date of the data record	—
`DelPro-13`	number	kg	—	Intake feed 1 from the milkingrobot	—
`DrgZetDat`	date	yy-mm-dd	Measured	Starting date of the dry period	—
`EiwStd`	number	%	Derived	Protein concentration of the monthly milk test	Milk crude protein percentage determined from official routine milk recording laboratory analysis. *Source:* `T_Animal_Milk Samples (MlkCon. ProteinPercentage)`

continued

Variable	Type	Unit	Origin	Description	Derivation / source
EndDatPrf	date	yy-mm-dd	Measured	Expected end date of the experiment	—
GewA	number	kg	Measured	Weight of the animal after milking in the afternoon	—
GewO	number	kg	Measured	Weight of the animal after milking in the morning	—
gf_visits	string	—	—	Measurements on cow emission (summarized data)	—
gf_cupdrops	string	—	—	GreenFeed portions concentrate feed.	—
Hok	integer	–	—	Location of where the cows are housed.	—
HokMin1	integer	–	—	Location of where the cows are housed day before	—
AtributedKVA	number	kg	Derived	Total allocated/programmed amount of concentrate feed type A for the cow on the specified date.	Sum of programmed concentrate A quantities across feeding stations (Hokofarm / SAC). *Source:* T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv
TotalIntakeKVA	number	kg	Derived	Total intake feed from SAC box 1	Total consumed quantity of concentrate feed A by the animal at the automatic feeding station on the specified date. *Source:* T_Raw_Hokofarm CD (totalIntakeKVA)
AtributedKVB	number	kg	Derived	Total allocated/programmed amount of concentrate feed type B for the cow on the specified date.	Sum of programmed concentrate B quantities across feeding stations (Hokofarm / SAC). *Source:* T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv
TotalIntakeKVB	number	kg	Derived	Total intake feed from SAC box 2	Total consumed quantity of concentrate feed B by the animal at the automatic feeding station on the specified date. *Source:* T_Raw_Hokofarm CD (totalIntakeKVB)
AtributedKVC	number	kg	Derived	Total allocated/programmed amount of concentrate feed type C for the cow on the specified date.	Sum of programmed concentrate C quantities across feeding stations (Hokofarm / SAC). *Source:* T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv
TotalIntakeKVC	number	kg	Derived	Total intake feed from SAC box 3	Total consumed quantity of concentrate feed C by the animal at the automatic feeding station on the specified date. *Source:* T_Raw_Hokofarm CD (totalIntakeKVC)
AtributedKVD	number	kg	Derived	Total allocated/programmed amount of concentrate feed type D for the cow on the specified date.	Sum of programmed concentrate D quantities across feeding stations (Hokofarm / SAC). *Source:* T_Raw_Hokofarm CD / TVL_Totaal_sub_ kv

continued

Variable	Type	Unit	Origin	Description	Derivation / source
TotalIntakeKVD	number	kg	Derived	Total intake feed from SAC box 4	Total consumed quantity of concentrate feed D by the animal at the automatic feeding station on the specified date. *Source:* T_Raw_HokofarmCD (totalIntakeKVD)
HrhNum	integer	–	—	Number of repeat of a previously performed test	—
InsDat	date	yy-mm-dd	Measured	Date of last insemination	—
Jar	integer	–	Measured	Year of the data record	—
KetStd	integer	–	Derived	Ketose concentration of the monthly milk test	Ketosis diagnostic classification code (0: normal/negative, 1: suspected subclinical, 2: clinical ketosis). *Source:* T_Animal_Milk Samples (MlkCon. KetosisCode)
KgMlkA	number	kg	Measured	Kilogram of milk registred at the afternoon milking event	—
KgMlkEtm	number	kg	Measured	Kilogram of milk in the last 24 hours	—
KgMlkO	number	kg	Measured	Kilogram of milk registred at the morning milking event	—
KlfDat	date	yy-mm-dd	Measured	Date of last calving	—
LacDgn	integer	–	Derived	Amount of days producing milk	Date - calving date
LacNum	integer	–	Measured	Parity (amount of calvings) of the animal	—
LacStd	number	%	Derived	Lactose of the monthly milk test	Milk lactose percentage determined from official routine milk recording laboratory analysis. *Source:* T_Animal_Milk Samples (MlkCon. LactosePercentage)
LevNum	integer	text + number	—	National registration number of the animal.	—
LftDgn	integer	–	Measured	Age of the animal in days	—
MlkStd	number	kg	Derived	Kilogram of milk produced during the standard (once a month) milk composition sampling	Total daily milk yield recorded during the official milk control recording day. *Source:* T_Animal_Milk Samples (MlkCon.Milk QuantityMass)
mpr_dag.Fat_ Concentration	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
mpr_dag.Lactose_ Concentration	number	%	Measured	Weekly milk lactose concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
mpr_dag.Protein_ Concentration	number	%	Measured	Weekly milk protein concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
mpr_dag.SCC	string	10^3 cells mL$^-$	Measured	Weekly somatic cell count during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
mpr_dag.Urea_ Concentration	string	mg dL^{-1}	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
mpr_week.Fat_ Concentration	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	ISO-calendar week yield-weighted average milk fat concentration calculated across regular CRV/MPR test recordings: sum(Fat_ Concentration * MilkingYield) / sum(MilkingYield). Used to derive Fat- and Protein-Corrected Milk (FPCM). *Source:* VL_Dier_Datum_ Crv
mpr_week.Lactose_ Concentration	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk lactose concentration calculated across regular CRV/MPR test recordings: sum(Lactose_ Concentration * MilkingYield) / sum(MilkingYield). *Source:* VL_Dier_Datum_ Crv
mpr_week.Protein_ Concentration	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk protein concentration calculated across regular CRV/MPR test recordings: sum(Protein_ Concentration * MilkingYield) / sum(MilkingYield). Used to derive Fat- and Protein-Corrected Milk (FPCM). *Source:* VL_Dier_Datum_ Crv
mpr_week.SCC	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average somatic cell count (1,000 cells/mL) calculated across regular CRV/MPR test recordings: sum(SCC * MilkingYield) / sum(MilkingYield). *Source:* VL_Dier_Datum_ Crv

continued

Variable	Type	Unit	Origin	Description	Derivation / source
mpr_week.Urea_Concentration	string	$\mathrm{mg\,dL^{-1}}$	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	ISO-calendar week yield-weighted average milk urea concentration (mg/dL) calculated across regular CRV/MPR test recordings: sum(Urea_Concentration * MilkingYield) / sum(MilkingYield). *Source:* VL_Dier_Datum_Crv
OorNum	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
OpnRvDsHans	string	—	—	Intake of dry matter from the roughage, Hans (Will be removed from table)	—
PrdSta	string	text	—	Identifying the stage of the cow, producing milk or not	—
PrfCod	integer	–	—	Code assigned by Dairy Campus to identify the experiment. Can be the same as the MyProjects number, or differ.	—
RowID	integer	–	—	Unique number for row in a specific table.	—
rv1.ds	string	—	—	Roughage 1 dry matter	—
rv1.name	string	text	—	Name of the roughage treatment consumed by the animal	—
rv1.opnDs	string	—	—	Roughage 1 intake dry matter	—
rv1.Prod	number	kg	Derived	Total uptake in kilograms of the roughage	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
rv1.ProdExclWater	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 1 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* VL_Totaal_sub_ric_opn_prod
rv1.ricBez	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
rv1.trBatches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
rv2.ds	number	kg	—	Roughage 2 dry matter	—
rv2.name	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
rv2.opnDs	number	kg	—	Roughage 2 intake dry matter	—
rv2.Prod	number	kg	Derived	Total uptake in kilograms of the roughage.	—
rv2.ProdExclWater	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 2 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* VL_Totaal_sub_ric_opn_prod

continued

Variable	Type	Unit	Origin	Description	Derivation / source
rv2.ricBez	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
rv2.trBatches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
rv3.ds	number	kg	—	Roughage 3 dry matter	—
rv3.name	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
rv3.opnDs	number	kg	—	Roughage 3 intake dry matter	—
rv3.Prod	number	kg	Derived	Total uptake in kilograms of the roughage.	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
rv3.ProdExclWater	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 3 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* VL_Totaal_sub_ric_opn_prod
rv3.ricBez	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
rv3.trBatches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—
rv4.ds	number	kg	—	Roughage 4 dry matter	—
rv4.name	string	text	—	Name of the roughage treatment consumed by the animal. These are only active (2, 3 and 4) when the animal consumes roughage from a treatment she isn't assigned to.	—
rv4.opnDs	number	kg	—	Roughage 4 intake dry matter	—
rv4.Prod	number	kg	Derived	Total uptake in kilograms of the roughage.	Sum of roughage intake of all visits in a single day. The roughage bin calculates uptake as: starting weight - end weight (bins contains a scale)
rv4.ProdExclWater	number	kg	Derived	Total uptake in kilograms of the roughage excluding water	Daily roughage intake fraction 4 excluding added water: (TotalLoaded - AddedWaterKg) / TotalLoaded * Intake. *Source:* VL_Totaal_sub_ric_opn_prod
rv4.ricBez	integer	–	Measured	Amount of times the animal visited the roughage bins to consume the roughage.	—
rv4.trBatches	integer	–	Measured	Amount of different batches distributed by the feeding robot that the animal consumed for its total roughage uptake	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
tot.opnDs	number	kg	Derived	Total uptake in kilograms of dry matter	Total daily dry matter intake (kg DM/day). Calculated by summing the dry matter intake of all roughage fractions: (OpnMinWater * DryMatterPercentage). *Source:* VL_Totaal_sub_ric_opn_prod
tot.Prod	number	kg	—	Total roughage 1 t/m 4	—
UreStd	number	mg/100 gram	Derived	Urea concentration of the monthly milk test	Milk urea nitrogen content determined from official routine milk recording laboratory analysis. *Source:* T_Animal_Milk Samples (MlkCon.Urea)
VetStd	number	%	Derived	Fat concentration of the monthly milk test	Milk fat percentage determined from official routine milk recording laboratory analysis. *Source:* T_Animal_Milk Samples (MlkCon.Fat Percentage)
VlgKlfDat	date	yy-mm-dd	Derived	Next calving date, not predicted, but added after calving	—
VrwDgn	integer	–	Derived	Number of days from date to expected calving date based on insemination date	Expected calving date minus current date. *Source:* TVL_Dier_ Datum_Stal
VrwKlfDat	date	yy-mm-dd	Derived	Expected calving date based on insemination date	Insemination date + 280 days
Wek	integer	–	Measured	Calender week number of data record	—
WekDag	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—

4.2 Automatic concentrate feeder transactions

Dataset: Hokofarm_SAC_concentrates | **Source table:** t_raw_hokofarmvb | **Variables:** 42

Table 3: Data dictionary of Hokofarm_SAC_concentrates.

Variable	Type	Unit	Origin	Description	Derivation / source
ashA	number	kg	Derived	Ash, indigestible part of feed A	—
ashB	number	kg	Derived	Ash, indigestible part of feed B	—
ashC	number	kg	Derived	Ash, indigestible part of feed C	—
ashD	number	kg	Derived	Ash, indigestible part of feed D	—
cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
crudeFiberA	number	kg	Derived	Crude fiber in feed A	—
crudeFiberB	number	kg	Derived	Crude fiber in feed B	—
crudeFiberC	number	kg	Derived	Crude fiber in feed C	—
crudeFiberD	number	kg	Derived	Crude fiber in feed D	—
date	date	yy-mm-dd	Measured	Date of the data record	—
dateTime	datetime	yyyy-mm-dd-hh-mm-ss	Measured	Date and time	—
dryMatterA	number	kg	Derived	Dry matter of feed A	—
dryMatterB	number	kg	Derived	Dry matter of feed B	—
dryMatterC	number	kg	Derived	Dry matter of feed C	—
dryMatterD	number	kg	Derived	Dry matter of feed D	—
endTime	number	hh-mm-ss	Measured	End time of action/behaviour	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
energyA	string	VEM	Derived	Energy content A	—
energyB	string	VEM	Derived	Energy content B	—
energyC	string	VEM	Derived	Energy content C	—
energyD	string	VEM	Derived	Energy content D	—
fatA	number	kg	Derived	Fat content A	—
fatB	number	kg	Derived	Fat content B	—
fatC	number	kg	Derived	Fat content C	—
fatD	number	kg	Derived	Fat content D	—
feedTypeA	string	text	—	Type of feed A	—
feedTypeB	string	text	—	Type of feed B	—
feedTypeC	string	text	—	Type of feed C	—
feedTypeD	string	text	—	Type of feed D	—
fileName	string	text	—	Name of a file	—
hashValue	integer	–	—	Validation record for data processing	—
intakeA	number	kg	Derived	Feed intake A	Calculated by endWeight - startWeight
intakeB	number	kg	Derived	Feed intake B	Calculated by endWeight - startWeight
intakeC	number	kg	Derived	Feed intake C	Calculated by endWeight - startWeight
intakeD	number	kg	Derived	Feed intake D	Calculated by endWeight - startWeight
proteinA	integer	–	Derived	Protein content A	—
proteinB	integer	–	Derived	Protein content B	—
proteinC	integer	–	Derived	Protein content C	—
proteinD	integer	–	Derived	Protein content D	—
startTime	number	hh-mm-ss	Measured	Start time of action/behaviour	—
station	integer	–	Measured	Station or location of activity	—
totalTime	integer	s	Derived	Total duration	Calculated by endTime - startTime
transponder	integer	–	Measured	Transponder number of the cow	—

4.3 Roughage intake control (RIC) visit transactions

Dataset: `Hokofarm_RIC` | **Source table:** `t_raw_hokofarmvr` | **Variables:** 18

Table 4: Data dictionary of `Hokofarm_RIC`.

Variable	Type	Unit	Origin	Description	Derivation / source
ash	number	kg	Derived	Ash, indigestible part of the feed	—
cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
crudeFibre	number	kg	Derived	Crude fiber in the feed	—
date	date	yy-mm-dd	Measured	Date of the data record	—
datetime	datetime	yyyy-mm-dd-hh-mm-ss	Measured	Date and time	—
dryMatter	number	kg	Derived	Dry matter	—
duration	integer	s	Derived	Duration of action/behaviour	Calculated by endTime - startTime
endTime	number	hh-mm-ss	Measured	End time of action/behaviour	—
endWeight	number	kg	Measured	End weight	—
energy	string	VEM	Derived	Energy content	—
fat	number	kg	Derived	Fat content	—
feedType	string	text	—	Type of feed	—
intake	number	kg	Derived	Feed intake	Calculated by endWeight - startWeight
protein	number	kg	Derived	Protein content	—
startTime	number	hh-mm-ss	Measured	Start time of action/behaviour	—
startWeight	number	kg	Measured	Start weight	—
station	integer	–	Measured	Station or location of activity	—
transponder	integer	–	Measured	Transponder number of the cow	—

4.4 Daily roughage intake and dry matter integration

Dataset: `TVL_RIC_intake_dm` | **Source table:** `TVL_totaal_hokofarm_ds` | **Variables:** 12

Table 5: Data dictionary of `TVL_RIC_intake_dm`.

Variable	Type	Unit	Origin	Description	Derivation / source
`cow`	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`date`	date	yy-mm-dd	Measured	Date of the data record	—
`dsPct`	integer	–	Derived	Percentage of dry matter in the feed	—
`feedtype`	string	text	—	Type of feed	—
`name`	string	text	—	Name collected based on other value	—
`opnDsVR`	number	kg	—	Intake of dry matter from roughage RIC	—
`opnIncWater`	number	kg	Derived	Feed intake including water	—
`opnMinWater`	number	kg	Derived	Feed intake excluding water	—
`opnProd`	number	kg	—	Total intake of product	—
`ricBez`	integer	–	—	Number of visits to RIC	—
`RowID`	integer	–	—	Unique number for row in a specific table.	—
`trBatches`	integer	–	—	Number of times the triolith has released feed.	—

4.5 Trioliet Triomatic feeding robot loading records

Dataset: `Triomatic_RIC_load` | **Source table:** `t_raw_trioliet` | **Variables:** 46

Table 6: Data dictionary of `Triomatic_RIC_load`.

Variable	Type	Unit	Origin	Description	Derivation / source
`AnimalCount`	integer	–	—	The number of animals	—
`Date`	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
`Extra`	integer	–	—	Extra information	—
`FeedingGroup`	integer	–	Measured	Log source 0 = 1, log source 1 = NULL betekenis Feeding group?	—
`filename`	string	text	—	Name of the file containing information	—
`FinishedReady`	string	—	—	NULL?	—
`GroupID`	integer	–	—	Id of the group	—
`GroupName`	string	text	—	Name of the group	—
`Id`	integer	–	—	ID	—
`IngredientName`	string	text	—	Name of the ingredient	—
`k42`	integer	–	—	Validation record for data processing	—
`LogSource`	integer	–	—	0 meaning feed preparation, and 1 the distribution of feed to RIC bins	—
`NettoLoadedWeight`	number	kg	Measured	Netto weight loaded	—
`NrOfSplittings`	integer	–	—	Number of splittings	—
`NumberRecipeLines`	integer	–	—	Total number of recipe lines	—
`OpdrachtID`	integer	–	—	Job ID	—
`PerformedFeedFactor`	integer	–	—	Performed feed factor	—
`PerformedProductDryWeightKgAfterDrop`	number	kg	Measured	Kg dry weight after drop	—
`PerformedProductDryWeightPartAfterDrop`	number	kg	Measured	Part of dry weight after drop	—
`PerformedProductError`	number	kg	Derived	Error, difference between Scheduled kg and performed kg feed	—
`PerformedProductKgAfterDrop`	number	kg	Measured	Kg product after drop	—
`PerformedProductKgAtStop`	number	kg	Measured	Kg product at stop	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
PerformedTime	number	hh-mm-ss	Measured	Performed time	—
PushingGroup	integer	–	—	Pushing group	—
Ready	integer	–	—	Ready	—
ReadyTime	number	hh-mm-ss	—	Ready time, if log source 1	—
ReceptLineNr	integer	–	—	Current number of recipe line	—
ReceptName	string	text	—	Name of the feed recipe	—
ScheduledDryWeight Content	integer	–	Derived	Scheduled dry weight content	—
ScheduledDryWeight ContentPart	integer	–	Derived	Scheduled dry weight content part	—
ScheduledDryWeightKg	number	kg	Derived	Scheduled dry weight kg	—
ScheduledFeedFactor	integer	–	—	Scheduled feed factor	—
ScheduledPart	integer	–	Derived	Scheduled part	—
ScheduledProductKg	number	kg	—	Scheduled product kg	—
ScheduledTime	number	hh-mm-ss	—	Scheduled time, if log source 1	—
SplittedNr	integer	–	—	Splitted number, only 1	—
StartPosition	integer	–	Measured	Start position, if subgroup not 0	—
StartTime	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
StartWeight	number	kg	Measured	Start weight, if subgroup not 0	—
StopPosition	integer	–	Measured	Stop position, if subgroup not 0	—
StopTime	number	hh-mm-ss	Measured	Stop time, if subgroup not 0	—
StopWeight	number	kg	Measured	Stop weight, if subgroup not 0	—
StrawGroup	integer	–	—	Straw group	—
SubGroupNr	integer	–	—	Number of the subgroup, 0 no subgroup	—
TakeoffTime	number	hh-mm-ss	Measured	Take off time, if log source 1	—
TrackSection	integer	–	—	Track section, if subgroup 1 if there is no subgroup 0	—

4.6 Trioliet Triomatic feeding robot trough deliveries

Dataset: Triomatic_RIC_drop | **Source table:** t_raw_trioliet | **Variables:** 46

Table 7: Data dictionary of Triomatic_RIC_drop.

Variable	Type	Unit	Origin	Description	Derivation / source
AnimalCount	integer	–	—	The number of animals	—
Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
Extra	integer	–	—	Extra information	—
FeedingGroup	integer	–	Measured	Log source 0 = 1, log source 1 = NULL betekenis Feeding group?	—
filename	string	text	—	Name of the file containing information	—
FinishedReady	string	—	—	NULL?	—
GroupID	integer	–	—	Id of the group	—
GroupName	string	text	—	Name of the group	—
Id	integer	–	—	ID	—
IngredientName	string	text	—	Name of the ingredient	—
k42	integer	–	—	Validation record for data processing	—
LogSource	integer	–	—	0 meaning feed preparation, and 1 the distribution of feed to RIC bins	—
NettoLoadedWeight	number	kg	Measured	Netto weight loaded	—
NrOfSplittings	integer	–	—	Number of splittings	—
NumberRecipeLines	integer	–	—	Total number of recipe lines	—
OpdrachtID	integer	–	—	Job ID	—
PerformedFeedFactor	integer	–	—	Performed feed factor	—
PerformedProductDry WeightKgAfterDrop	number	kg	Measured	Kg dry weight after drop	—
PerformedProductDry WeightPartAfterDrop	number	kg	Measured	Part of dry weight after drop	—
PerformedProductError	number	kg	Derived	Error, difference between Scheduled kg and performed kg feed	—
PerformedProductKg AfterDrop	number	kg	Measured	Kg product after drop	—
PerformedProductKgAt Stop	number	kg	Measured	Kg product at stop	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
PerformedTime	number	hh-mm-ss	Measured	Performed time	—
PushingGroup	integer	–	—	Pushing group	—
Ready	integer	–	—	Ready	—
ReadyTime	number	hh-mm-ss	—	Ready time, if log source 1	—
ReceptLineNr	integer	–	—	Current number of recipe line	—
ReceptName	string	text	—	Name of the feed recipe	—
ScheduledDryWeight Content	integer	–	Derived	Scheduled dry weight content	—
ScheduledDryWeight ContentPart	integer	–	Derived	Scheduled dry weight content part	—
ScheduledDryWeightKg	number	kg	Derived	Scheduled dry weight kg	—
ScheduledFeedFactor	integer	–	—	Scheduled feed factor	—
ScheduledPart	integer	–	Derived	Scheduled part	—
ScheduledProductKg	number	kg	—	Scheduled product kg	—
ScheduledTime	number	hh-mm-ss	—	Scheduled time, if log source 1	—
SplittedNr	integer	–	—	Splitted number, only 1	—
StartPosition	integer	–	Measured	Start position, if subgroup not 0	—
StartTime	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
StartWeight	number	kg	Measured	Start weight, if subgroup not 0	—
StopPosition	integer	–	Measured	Stop position, if subgroup not 0	—
StopTime	number	hh-mm-ss	Measured	Stop time, if subgroup not 0	—
StopWeight	number	kg	Measured	Stop weight, if subgroup not 0	—
StrawGroup	integer	–	—	Straw group	—
SubGroupNr	integer	–	—	Number of the subgroup, 0 no subgroup	—
TakeoffTime	number	hh-mm-ss	Measured	Take off time, if log source 1	—
TrackSection	integer	–	—	Track section, if subgroup 1 if there is no subgroup 0	—

4.7 GEA separation and smart gate passages

Dataset: GEA_gates | **Source table:** T_RAW_Gea_Separatie | **Variables:** 5

Table 8: Data dictionary of GEA_gates.

Variable	Type	Unit	Origin	Description	Derivation / source
cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
date	date	yy-mm-dd	Measured	Date of the data record	—
event	integer	–	—	0 = No gate action or weighting, 1 = selection gate A, 2 = selection gate B	—
gate	integer	–	—	Number of the gate	—
message	string	tekst	—	ID and info gate action, weighting or selection has or has not taken place	—

4.8 GEA milking transactions

Dataset: GEA_milkings | **Source table:** T_RAW_Gea_milk | **Variables:** 13

Table 9: Data dictionary of GEA_milkings.

Variable	Type	Unit	Origin	Description	Derivation / source
Cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
IMDat	datetime	yy-mm-dd-hh-mm-ss	—	GEA, Date and time milking	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
Reg1	integer	text + number	—	National registration number of the animal.	—
RespISO	integer	–	—	Number of the cows responder	—
Name	string	—	—	Number in GEA, should be same as cow	—
IMTim	string	—	—	Time?	—
IMWt	number	kg	Measured	Live animal body weight measured by the integrated scale inside the GEA milking stall or exit platform.	Directly recorded by GEA milking scale sensor. *Source:* T_RAW_Gea_milk
IMFlg	string	code	Measured	Status/diagnostic flag from GEA milking robot/parlor indicating normal completion, kick-off, or incomplete milking.	Status code recorded by GEA system. *Source:* T_RAW_Gea_milk
IMDim	integer	days	Derived	Days in Milk (DIM) recorded in GEA at the time of the milking event.	Milking date minus most recent calving date. *Source:* T_RAW_Gea_milk
IMAvFlo	string	—	—	Average milkflow?	—
IMMxFlow	string	—	—	Max. milkflowAlleen '||||' als waarde???	—
IMStall	integer	–	—	Place in the Gea milking carousel 1-40	—
IMStartSec	string	—	—	Alleen '?????' als waarde???	—

4.9 CowConnect mixer wagon rations

Dataset: TMR_Trioliet_cowconnect_mix | **Source table:** t_raw_cowconnect_mix | **Variables:** 12

Table 10: Data dictionary of TMR_Trioliet_cowconnect_mix.

Variable	Type	Unit	Origin	Description	Derivation / source
date	date	yy-mm-dd	Measured	Date of the data record	—
externalFeedplanID	string	—	—	All NULL	—
feedplanId	integer	–	—	Id of the feedplan	—
feedplanName	string	text	—	Name of the feedplan	—
id	string	text	—	ID	—
isOpen	integer	–	—	1 or 0 process running or not.	—
loadedSum	number	kg	Derived	Total / sum of all loads	—
mixDate	datetime	yy-mm-dd-hh-mm-ss	Measured	Date and time	—
mixLoaderId	integer	–	—	ID of the mix loader	—
mixLoaderName	string	text	—	Name of the mix loader	—
rationId	string	—	—	All NULL	—
type	string	text	—	Type	—

4.10 Nedap Velos behaviour intervals

Dataset: Velos_behaviour | **Source table:** T_RAW_velos_behaviour | **Variables:** 4

Table 11: Data dictionary of Velos_behaviour.

Variable	Type	Unit	Origin	Description	Derivation / source
date	date	yy-mm-dd	Measured	Date of the data record, UTC timezone	—
cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
type	string	text	—	Type of behaviour 1 - 3	0, overig actief. 1, niet actief. 2, herkouwen. 3, vreten.
duration	integer	s	Derived	Duration of action/behaviour	Calculated by endTime - startTime

4.11 Nedap Velos daily behaviour budgets

Dataset: `Velos_behaviour_day` | **Source table:** `T_RAW_velos_behaviour` | **Variables:** 7

Table 12: Data dictionary of `Velos_behaviour_day`.

Variable	Type	Unit	Origin	Description	Derivation / source
`date`	date	yyyy-mm-dd	Measured	Date of the data record, UTC timezone	—
`cow`	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`total`	number	hh-mm-ss	Derived	Total time recorded of the animal 1 day	Sum duration of all types *Source:* `T_RAW_velos_behaviour`
`overig`	number	hh-mm-ss	Derived	Time the cow is actief but not eating or ruminating	Sum duration of type 0 *Source:* `T_RAW_velos_behaviour`
`nietactief`	number	hh-mm-ss	Derived	Time the cow is inactief	Sum duration of type 1 *Source:* `T_RAW_velos_behaviour`
`herkouwen`	number	hh-mm-ss	Derived	Time the cow spent ruminating	Sum duration of type 2 *Source:* `T_RAW_velos_behaviour`
`vreten`	number	hh-mm-ss	Derived	Time the cow spent eating	Sum duration of type 3 *Source:* `T_RAW_velos_behaviour`

4.12 CowConnect feed distribution drops

Dataset: `TMR_Trioliet_cowconnect_drop` | **Source table:** `t_raw_cowconnect_drop` | **Variables:** 11

Table 13: Data dictionary of `TMR_Trioliet_cowconnect_drop`.

Variable	Type	Unit	Origin	Description	Derivation / source
`id`	string	text	—	ID	—
`externalMixTaskId`	string	text	—	External mix task ID	—
`quantity`	number	kg	Measured	Quantity	—
`leftOverQuantity`	number	kg	Measured	Quantity left over	—
`noOfAnimals`	integer	–	—	Number of animals	—
`feedSectionName`	string	text	—	Name of the feed section	—
`feedsectionType`	integer	–	—	Feedsection type	—
`externalFeedSectionId`	integer	–	—	External Feed section ID if the feed section has an external ID	—
`externalMixTask`	string	—	—	—	—
`mix_id`	string	text	—	Mix ID and external mix task id are foreign keys of the ID (primairy key)	—
`type`	string	text	—	Type	—

4.13 Laboratory dry matter content (drying oven)

Dataset: `DC_DM_content` | **Source table:** `VL_totaal_droogstoof` | **Variables:** 16

Table 14: Data dictionary of `DC_DM_content`.

Variable	Type	Unit	Origin	Description	Derivation / source
`BlkDat`	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the can was weighed	—
`BlkGew`	number	g	Measured	Weight of the can	—
`BlkNum`	integer	–	—	Number of the can	—
`DatMon`	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
DatSes	datetime	yy-mm-dd-hh-mm-ss	Measured	Date of weigh session	—
DrgDat	datetime	yy-mm-dd-hh-mm-ss	Measured	Date end of the analysis	—
DrgGew	number	g	Measured	Dried sample weight, with or without weight of the can?	—
DrgStfGkg	number	g	Measured	Weight of the dry matter	—
DrgTyd	number	hh-mm-ss	Measured	Time end of the analysis when all moisture has evaporated	—
ObjCod	string	text	—	Short description/code of the sample	—
ObjCodNum	integer	–	—	Used when analyzing duplicate or triplicate samples	—
PrjCod	string	text	—	Project code	—
VlgNum	integer	–	—	Tracking number	—
VrsDat	datetime	yy-mm-dd-hh-mm-ss	Measured	Date start of the analysis	—
VrsGew	number	g	Measured	Weight at the start of the analysis	—
VrsTyd	number	hh-mm-ss	Measured	Time start of the analysis, fresh sample is weighed	—

4.14 GreenFeed enteric emission summaries

Dataset: Greenfeed_summarized_data | **Source table:** T_RAW_Greenfeed_Summarized_Data | **Variables:** 24

Table 15: Data dictionary of Greenfeed_summarized_data.

Variable	Type	Unit	Origin	Description	Derivation / source
AirflowCf	integer	–	—	Airflow correction factor	—
AirflowLitersPerSec	integer	$\mathrm{L\,s^{-1}}$	Measured	Airflow liters per second	—
AnimalName	integer	–	Measured	Animal name	—
CH4GramsPerDay	integer	–	Measured	CH4 grams per day, methane	—
CO2GramsPerDay	integer	–	Measured	CO2 grams per day, carbon dioxide	—
cow	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
date	date	yy-mm-dd	Measured	Date of the data record	—
EndTime	number	yy-mm-dd-hh-mm-ss	Measured	End time	—
GoodDataDuration	datetime	hh-mm-ss	—	Good data duration	—
greenfeed	integer	–	Measured	Greenfeed	—
H2GramsPerDay	integer	–	Measured	H2 grams per day, hydrogen	—
H2SGramsPerDay	integer	–	Measured	H2S grams per day, hydrogen sulfide	—
InterruptingTags	integer	–	—	Interrupting tags	—
IsPreliminary	integer	–	—	Is preliminary data, validated data comes directly from C-lock.	—
O2GramsPerDay	integer	–	Measured	O2 grams per day, oxygen	—
RFID	integer	–	—	Rfid	—
RunTime	number	yy-mm-dd-hh-mm-ss	Measured	Run time	—
StartTime	number	hh-mm-ss	Measured	Start time, if subgroup not 0	—
TempPipeDegreesCelsius	number	°C	Measured	Temperature pipe degrees celsius	—
user	string	text	—	User	—
WasInterrupted	integer	–	—	Was interrupted	—
WindCf	integer	–	—	Wind correction factor	—
WindDirDeg	integer	–	Measured	Wind direction degrees	—
WindSpeedMetersPerSec	integer	$\mathrm{m\,s^{-1}}$	Measured	Wind speed meters per second	—

4.15 GreenFeed bait feed drops

Dataset: Greenfeed_feed_aggregrated | **Source table:** T_RAW_Greenfeed_Feed_data | **Variables:** 13

Table 16: Data dictionary of `Greenfeed_feed_aggregrated`.

Variable	Type	Unit	Origin	Description	Derivation / source
`cow`	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number. Same as AnimalNo, is unique per date.	—
`CurrentCup`	integer	–	Measured	Current cup	—
`CurrentPeriod`	integer	–	Measured	Current period	—
`date`	date	yy-mm-dd	Measured	Date of the data record	—
`DateTime`	datetime	yyyy-mm-dd hh-mm-ss	Measured	Date and time	—
`DropInterval`	integer	s	—	Min. sec between current en new cup	—
`FeedType`	integer	–	—	Feed type	—
`greenfeed`	integer	–	Measured	Greenfeed	—
`MaxCups`	integer	–	—	Max. cups per period	—
`MaxPeriods`	integer	–	—	Max. periods per day	—
`PeriodInterval`	integer	s	—	Min. Sec between current en next period	—
`RFIDTag`	integer	–	Measured	Rfid tag	—
`user`	string	text	—	User	—

4.16 Dairy Campus meteorological station

Dataset: `Weatherstation` | **Source table:** `T_RAW_weerstation` | **Variables:** 18

Table 17: Data dictionary of `Weatherstation`.

Variable	Type	Unit	Origin	Description	Derivation / source
`dewpoint`	number	°C	Measured	Dew point is the temperature to which air must be cooled to become saturated with water vapor	—
`global_radiation`	string	$\mathrm{W\,m^{-2}}$	Measured	The sum of the diffuse and direct solar radiation is called global solar radiation.	—
`gust`	integer	Beaufort, over 10 minuten (meter/sec)	Measured	A sudden, brief increase in speed of the wind	—
`gust_direction`	string	°	Measured	Direction from which the wind gust blows	—
`humidity_150`	number	%	Measured	The concentration of water vapor present in the air	—
`latitude`	string	°	Measured	Coordinate that specifies the north–south position	—
`longitude`	string	°	Measured	Coordinate that specifies the east–west position	—
`precipitation`	string	Millimeters	Measured	Any product of the condensation of atmospheric water vapor that falls from clouds	—
`rain_chance`	number	%	Measured	Chance of precipitation	—
`station_name`	string	text	—	Station name	—
`symbol_number`	string	—	—	Symbol number	—
`temperature_10`	number	°C	Measured	Temperature 10 cm above ground level.	—
`temperature_150`	number	°C	Measured	Temperature 150 cm above ground level.	—
`timestamp`	datetime	yy-mm-dd-hh-mm-ss	Measured	Timestamp	—
`wet_bulb_temperature`	number	°C	Measured	Thermometer with the bulb in somthing wet	—
`wind_direction`	string	°	Measured	Direction from which the wind blows	—
`windchill`	number	°C	Measured	The cooling effect of wind blowing on a surface	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
windspeed	integer	Beaufort, over 10 minuten (meter/sec)	Measured	Rate at which wind/air is moving	—

4.17 Data completeness audit matrix

Dataset: Missing_data | **Source table:** tvl_totaal_missende_data_proef | **Variables:** 17

Table 18: Data dictionary of Missing_data.

Variable	Type	Unit	Origin	Description	Derivation / source
bcs	boolean	yes/no	Measured	Automatic body condition score camera	number
behcod	string	yes/no	Measured	Treatment code	—
bijzonderheden	boolean	yes/no	Measured	Particularities	—
date	date	yy-mm-dd	Measured	Date of the data record	—
gewa	boolean	yes/no	Measured	Weight evening	kg
gewgeaa	boolean	yes/no	Measured	Weight gea evening	kg
gewgeao	boolean	yes/no	Measured	Weight gea morning	kg
gewo	boolean	yes/no	Measured	Weight morning	kg
hok	boolean	yes/no	Measured	Barn	number
kgmlka	boolean	yes/no	Measured	Kg milk evening	kg
kgmlko	boolean	yes/no	Measured	Kg milk morning	kg
oornum	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
opnkv	boolean	yes/no	Measured	Intake feed	kg
opnrv	string	yes/no	Measured	Intake roughage	kg
prdsta	boolean	yes/no	Measured	Identifying the stage of the cow, producing milk or not	short text
prfcod	string	yes/no	Measured	Trial code	—
velos	boolean	yes/no	Measured	Velos cow behavior	—

4.18 Barn environment and animal daily profile

Dataset: Environment_barn_Animal_day | **Source table:** TVL_Dier_Datum_Stal | **Variables:** 52

Table 19: Data dictionary of Environment_barn_Animal_day.

Variable	Type	Unit	Origin	Description	Derivation / source
AgeDAys	integer	–	Measured	Age of the animal in days	—
AnimalNumber	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
AntGewA	number	kg	Measured	Weight of the animal after milking in the afternoon	—
AntGewO	number	kg	Measured	Weight of the animal after milking in the morning	—
BirthDate	date	yyyy-mm-dd	Measured	The cows date of birth	—
DateMelkCon4Wk	date	yyyy-mm-dd	Measured	Date of the most recent official 4-weekly milk control / recording event (MPR).	Date of official test day sampling. *Source:* T_Animal_Milk Samples
DatGldTot	datetime	yyyy-mm-dd	Derived	Date until which the current housing, lactation, or trial assignment status remains valid.	End validity date of status record. *Source:* TVL_Dier_ Datum_Stal
DaysInLactation	integer	–	Derived	Amount of days producing milk	Date - calving date
DryDate	date	yy-mm-dd	Measured	Starting date of the dry period	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
EarNumber	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
Eiw	number	%	Derived	Protein concentration of the monthly milk test	—
EndDate	date	yy-mm-dd	Measured	Expected end date of the experiment	—
Ket	string	Calculated	Derived	number	Laboratory diagnostic score from milk recording analysis. *Source:* T_Animal_Milk Samples
ExpecCalvingDate	date	yy-mm-dd	Derived	Expected calving date based on insemination date	Calculated by ExpectedCalvingDate() function from last successful insemination. *Source:* t_animal_ matings / Expected CalvingDate
Fat%	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	—
GewTot	number	kg	Measured	Average weight of the animal for the day.	Recorded by automated walkthrough weighing scale. *Source:* TVL_Dier_ Gewicht_Etmaal
IDNumber	integer	National registration number of the animal.	Measured	Registration number same as LevNum	Official national cattle ear tag number. *Source:* t_animals
InsiminationDate	date	yy-mm-dd	Measured	Date of last insemination	—
Lac	number	%	Derived	Lactose of the monthly milk test	—
Lactose%	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	—
LastCalvingDate	date	yy-mm-dd	Measured	Date of last calving	—
Milk_Yield	number	kg	Derived	Milk yield	—
MilkAfternoon	number	kg	Derived	Milk Yield afternoon	—
MilkDay	number	kg	Derived	Milk Yield day	—
MilkMorning	number	kg	Derived	Milk Yield morning	—
MilkYield	number	kg	Derived	Milk yield	—
MlkCon	boolean	boolean	Derived	Indicator flag showing whether official milk recording (MPR) data was collected on this date.	Boolean flag indicating presence of milk laboratory sample record. *Source:* T_Animal_Milk Samples
NumberAnimal	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
NumberOLactations	integer	–	Measured	Parity (amount of calvings) of the animal	—
NumbOfDates	date	count	Derived	Count of observation days recorded for the animal within the experimental reporting period.	Count of distinct active recording dates. *Source:* TVL_Dier_ Datum_Proef
Protein%	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	—
SampleDateEvening	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the evening	—
SampleDateMorning	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the morning	—
SCC	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
StartDate	datetime	hh-mm-ss	Measured	Start date	—
Stl	string	identifier	Measured	Physical barn building or housing section identifier at Dairy Campus.	Stable / Barn number recorded in management system. *Source:* TVL_Dier_Datum_Stal
SumVoeGft	number	kg	Derived	Total amount of concentrate feed programmed/granted to the animal over the period.	Sum of programmed concentrate feeding amounts. *Source:* TVL_Dier_Voeropname_KV_Etmaal
SumVoeOpnm	number	kg	Derived	Total actual concentrate feed consumed by the animal over the period.	Sum of consumed concentrate intake amounts. *Source:* TVL_Dier_Voeropname_KV_Etmaal
SumVoeRst	number	kg	Derived	Total amount of unconsumed concentrate feed remaining in the troughs.	Total Feed Granted minus Total Feed Intake. *Source:* TVL_Dier_Voeropname_KV_Etmaal
Unit	integer	–	—	Location of where the cows are housed.	—
Ure	string	$\mathrm{mg\,dL^{-1}}$	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	—
Urea (mg/100g)	string	$\mathrm{mg\,dL^{-1}}$	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
Vet	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
Week	integer	–	Measured	Calender week number of data record	—
WeekDay	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—
WeightAfternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
WeightMorning	number	kg	Measured	Weight of the animal after milking in the morning	—
Year	string	—	—	Measured	—
VrwDgn	integer	days	Derived	Expected remaining gestation days or planned days dry prior to the anticipated calving date.	Expected calving date minus current date. *Source:* TVL_Dier_Datum_Stal
AntMlkO	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
AntMlkA	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—

4.19 Experimental cohort animal daily records

Dataset: Environment_barn_Animal_experim | **Variables:** 54

Table 20: Data dictionary of Environment_barn_Animal_experim.

Variable	Type	Unit	Origin	Description	Derivation / source
AntMlkA	integer	–	Measured	How often an animal was milked in the afternoon. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
AntMlkO	integer	–	Measured	How often an animal was milked in the morning. 1 is the assumed value. 0 means a missing value/something wrong with registration. 2+ means reattachment of the milking apparatus, can be outlier values.	—
Date	datetime	yy-mm-dd-hh-mm-ss	Measured	Date	—
VrwDgn	integer	days	Derived	Expected remaining gestation days or planned days dry prior to the anticipated calving date.	Expected calving date minus current date. *Source:* TVL_Dier_Datum_Stal
AgeDays	integer	days	Derived	Age of the animal in days calculated as recording date minus birth date.	Recording date - Birth date. *Source:* TVL_Dier_Datum_Stal
AnimalNumber	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
BirthDate	date	yyyy-mm-dd	Measured	The cows date of birth	—
DaysInLactation	integer	–	Derived	Amount of days producing milk	Date - calving date
DryDate	date	yy-mm-dd	Measured	Starting date of the dry period	—
EarNumber	integer	–	—	Number on the ear tag of the animal, length 3-4 numbers. Derives from the registration number.	—
EndDate	date	yy-mm-dd	Measured	Expected end date of the experiment	—
Fat%	number	%	Derived	An estimated daily milk fat concentration based on the AO sampling and the corresponding milk yield of that day. x	—
GewTot	number	kg	Measured	Total live animal body weight (average or latest measurement for the date).	Recorded by automated walkthrough weighing scale. *Source:* TVL_Dier_Gewicht_Etmaal
InsiminationDate	date	yy-mm-dd	Measured	Date of last insemination	—
Ket	string	code	Derived	Ketosis classification indicator (0: healthy/negative, 1: suspected subclinical ketosis, 2: clinical ketosis).	Laboratory diagnostic score from milk recording analysis. *Source:* T_Animal_Milk Samples
Lactose%	number	%	Derived	An estimated daily milk lactose concentration based on the AO sampling and the corresponding milk yield of that day.	—
LastCalvingDate	date	yy-mm-dd	Measured	Date of last calving	—

continued

Variable	Type	Unit	Origin	Description	Derivation / source
NumberAnimal	integer	–	—	Unique animal server identifier issued across the different systems at Dairy Campus	—
NumberOLactations	integer	–	Measured	Parity (amount of calvings) of the animal	—
NumbOfDates	date	count	Derived	Count of observation days recorded for the animal within the experimental reporting period.	Count of distinct active recording dates. *Source:* TVL_Dier_Datum_Proef
Projectcode	string	identifier	Measured	Unique research experiment / trial code (e.g. 4400003634). Same as prfcod.	Official trial identifier registered in t_proef. *Source:* t_proef
ProjectName	string	text	Measured	Official descriptive name of the research project or trial. Same as prfnam.	Official trial title registered in t_proef. *Source:* t_proef
Protein%	number	%	Derived	An estimated daily milk protein concentration based on the AO sampling and the corresponding milk yield of that day.	—
SampleDateEvening	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the evening	—
SampleDateMorning	datetime	yy-mm-dd-hh-mm-ss	Measured	Date the sample was obtained during the morning	—
SCC	string	10^3 cells mL$^-$	Derived	An estimated daily somatic cell count based on the AO sampling and the corresponding milk yield of that day.	—
StartDate	datetime	hh-mm-ss	Measured	Start date	—
SumVoeGft	number	kg	Derived	Total amount of concentrate feed programmed/granted to the animal over the period.	Sum of programmed concentrate feeding amounts. *Source:* TVL_Dier_Voeropname_KV_Etmaal
SumVoeRst	number	kg	Derived	Total amount of unconsumed concentrate feed remaining in the troughs.	Total Feed Granted minus Total Feed Intake. *Source:* TVL_Dier_Voeropname_KV_Etmaal
Unit	integer	–	—	Location of where the cows are housed.	—
Urea (mg/100g)	string	mg dL^{-1}	Measured	Weekly milk urea concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
Week	integer	–	Measured	Calender week number of data record	—
WeekDay	string	text	—	Abbreviaton of weekday (Ma = Monday, Di = Tuesday, Wo = Wednesday, Do = Thursday, Vr = Friday, Za = Saturday, Zo = Sunday)	—
WeightAfternoon	number	kg	Measured	Weight of the animal after milking in the afternoon	—
WeightMorning	number	kg	Measured	Weight of the animal after milking in the morning	—
Year	string	—	—	Measured	—
AntGewA	number	kg	Measured	Weight of the animal after milking in the afternoon	—
AntGewO	number	kg	Measured	Weight of the animal after milking in the morning	—
IDNumber	string	identifier	Measured	Official national unique life identification number of the animal (LevNum / EU identification tag).	Official national cattle ear tag number. *Source:* t_animals
DateMelkCon4Wk	date	yyyy-mm-dd	Measured	Date of the most recent official 4-weekly milk control / recording event (MPR).	Date of official test day sampling. *Source:* T_Animal_Milk Samples

continued

Variable	Type	Unit	Origin	Description	Derivation / source
DatGldTot	datetime	yyyy-mm-dd	Derived	Date until which the current housing, lactation, or trial assignment status remains valid.	End validity date of status record. *Source:* TVL_Dier_Datum_Stal
Eiw	number	%	Derived	Protein concentration of the monthly milk test	—
ExpecCalvingDate	date	yyyy-mm-dd	Derived	Estimated date of the next calving, based on confirmed insemination date + gestation length (~270-280 days).	Calculated by ExpectedCalvingDate() function from last successful insemination. *Source:* t_animal_matings / ExpectedCalvingDate
Lac	number	%	Derived	Lactose of the monthly milk test	—
Milk_Yield	number	kg	Derived	Milk yield	—
MilkAfternoon	number	kg	Derived	Milk Yield afternoon	—
MilkDay	number	kg	Derived	Milk Yield day	—
MilkMorming	number	kg	Measured	Total quantity of milk produced during morning milking sessions.	Sum of morning milking yields. *Source:* TVL_Dier_Melkgift_Etmaal_Melkcontrole
MilkYield	number	kg	Derived	Milk yield	—
MlkCon	boolean	boolean	Derived	Indicator flag showing whether official milk recording (MPR) data was collected on this date.	Boolean flag indicating presence of milk laboratory sample record. *Source:* T_Animal_Milk Samples
Stl	string	identifier	Measured	Physical barn building or housing section identifier at Dairy Campus.	Stable / Barn number recorded in management system. *Source:* TVL_Dier_Datum_Stal
Ure	string	$\mathrm{mg\,dL^{-1}}$	Derived	An estimated daily milk urea concentration based on the AO sampling and the corresponding milk yield of that day.	—
Vet	number	%	Measured	Weekly milk fat concentration during the AO sampling for animals in an experiment. AO = avond ochtend (eveningn morning) sample, taken on Tuesday evening and Wednesday morning. So 2 values per week.	—
SumVoeOpnm	number	kg	Derived	Total actual concentrate feed consumed by the animal over the period.	Sum of consumed concentrate intake amounts. *Source:* TVL_Dier_Voeropname_KV_Etmaal

5 Data availability and citation

The machine-readable dictionary (CSV, XLSX), per-dataset CSV files and the SQL calculation layer (views, stored procedures and functions) are deposited with this descriptor. Please cite:

> Klandermans, B., & de Jong, F. (2026). *Dairy Campus Research Data Platform: Data Descriptor, Standardised Metadata Dictionaries and Calculation Methodology.* Zenodo. https://doi.org/10.5281/zenodo.24092174